Guru Nanak Institute of Pharmaceutical Science and Technology.

Name:- PRAMIT SAHOO

Roll no.:- 1860191901 3

Sec:- A

Year:- 3rd Year (5th Sem)

Stream:- B.Pharm

Reg. No:- 030615 OF 2019-2020

Sub:- Pharmacology-II practical

Sub Code:- PT598

1. **Bioassay.** –

It is a analytical method to determine concentration or potency of a substance by its effect on living cells or tissues.

Types:- qualitative bioassay:

It is used for assessing the physical effects of a substance that may not be quantified. such as abnormal development.

Quantitative bioassay:- It involves estimation of concentration potency of a substance by measurement of the biological response it produces.

Application of bioassay:

→ To measure the pharmacological activity of new chemically unidentified substances.

→ To investigate the function of endogenous mediators

→ To measure drug toxicity and unwanted effects.

Scanned with CamScanner

→ To measure the concentration of drugs and other active substances in the blood or other body fluids.

Procedure:

1. open the software page.
2. select the particular experiment.
3. we selected the drug which need to be injected.
4. we select particular dose (i.e → 0.1, 0.2, 0.4, ...)
5. observe the response at kymograph
6. Then, consider a dose the standard dose.
7. The we have to calculate the response of unknown dose that is matching with standard dose.
8. Now select unknown to inject.
9. The conc. of the unknown is calculated using formula depending on the type of bioassay.

Scanned with CamScanner

2. (i) **Mechanism of carageenan:**

The main mechanism of action of NSAIDS like carageenan is the inhibition of enzyme cyclooxygenase (COX). It is req. to convert arachidonic acid into thromboxanes / prostaglandins, and prostacyclins. The therapeutic effect of NSAIDS are attributed to the lack of these eicosanoids. Specifically throboxanes play a role in platelet adehsion.

2. (ii) **PA2 value :-**

pA2 is a measure of the potency of an antagonist. It is the negative logarithm of the molar concentration of an antagonist that produce a twofold shift in the conc. response curve for an antagonist.

pA2 value determines the the conc. of antagonist when double the agonist is required to have the same effect.

The meaning of pA_2 is the affinity of the antagonist to the receptor.

2. (iii) **Calculation of unknown concentration by 4 point bioassay:-**

The formula for calculating unknown conc. is.

$$Mt_1 = \left(\frac{n_1}{t_1}\right) \times \text{antilog}\left[\frac{(S_1+S_2)-(T_1+T_2)}{(S_2+T_2)-(S_1+T_1)}\right] \times \log\left(\frac{n_2}{n_1}\right)$$

where, $n_1 \Rightarrow$ lower standard dose
$n_2 \rightarrow$ higher " "
$S_1 \rightarrow$ Response of n_1
$S_2 \rightarrow$ " " n_2

$Mt_1 \rightarrow$ Lower conc. of test sample
$t_1 \rightarrow$ vol of lower test sample
$t_2 \rightarrow$ " " higher " "
$T_1 \rightarrow$ Response of t_1
$T_2 \rightarrow$ " " t_2

Scanned with CamScanner

2.(iv) Effect of Isoprenaline on systolic BP:-

Isoprenaline is a non-selective β-adrenergic agonist. It has positive Inotropic and chronotropic effects, increasing cardiac output by increasing the heart rate and cardiac contractility. Isoprenaline also decreases diastolic blood pressure by lowering peripheral vascular resistance.

2.(v) Normal heart rate value of dog is → 120 to 160 beats per minute.

Scanned with CamScanner